Software Engineering

CEN-220

Project Documentation

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SMART TRAFFIC MANAGEMENT SYSTEM (STMS)

MILESTONE 1: REQUIREMENTS

1. INTRODUCTION

The Smart Traffic Management System (STMS) is an AI and IoT-based intelligent system designed to monitor and control traffic flow in real time. It uses sensors, cameras, and AI-based decision-making to dynamically adjust traffic signals, reduce congestion, and prioritize emergency vehicles.

The system improves:

- Traffic efficiency
- Emergency response time
- Urban mobility
- Road safety

2. PROBLEM STATEMENT

Modern cities face serious traffic congestion problems due to:

- Poor signal timing
- Manual traffic control
- Emergency vehicle delays
- Lack of real-time monitoring
- High accident risk
- Traffic jams during peak hours

Traditional traffic systems cannot make fast decisions based on live traffic conditions.

The Smart Traffic Management System solves this problem using:

- AI-based traffic analysis
- Smart sensors
- Automated signal control
- Emergency vehicle priority
- Real-time monitoring dashboard

This improves:

- Traffic flow

- Road safety
- Emergency response time
- Fuel efficiency
- City management

3. PROJECT SCOPE

The Smart Traffic Management System (STMS) project covers the following areas:

- Monitor traffic flow at multiple road intersections in real time.
- Collect traffic data using smart sensors, cameras, and detectors.
- Analyze traffic density and identify congestion levels automatically.
- Adjust traffic signal timing based on live traffic conditions.
- Detect emergency vehicles such as ambulances, fire brigades, and police vehicles.
- Provide emergency vehicle priority by clearing traffic routes.
- Allow traffic police and admin to manually override traffic signals when needed.
- Display live traffic conditions through a monitoring dashboard.
- Generate alerts for congestion, accidents, signal failures, and sensor malfunctions.
- Store traffic records for future analysis and reporting.
- Generate daily, weekly, and monthly traffic performance reports.
- Support multiple intersections in a city-wide traffic network.
- Improve road safety and reduce waiting time for vehicles.
- Optimize fuel usage by reducing unnecessary traffic delays.
- Provide historical traffic data for future planning and decision making.

4. FUNTIONAL REQUIREMENTS

1. Traffic Data Collection

- The system shall collect real-time traffic data from road sensors and cameras.
- The system shall detect vehicle count on each lane.
- The system shall monitor traffic speed and movement patterns.

2. Traffic Analysis

- The system shall analyze traffic density continuously.
- The system shall classify traffic levels as low, medium, or high.
- The system shall identify traffic congestion points.

3. Signal Control

- The system shall dynamically adjust traffic signal timing.
- The system shall extend green signal duration during heavy traffic.
- The system shall reduce signal time during low traffic.
- The system shall coordinate signal synchronization between connected intersections.

4. Emergency Vehicle Handling

- The system shall detect emergency vehicle signals instantly.
- The system shall override normal traffic signal operations during emergencies.
- The system shall create a clear traffic path for emergency vehicles.
- The system shall restore normal traffic operation after emergency passage.

5. Monitoring Dashboard

- The system shall display live traffic conditions.
- The system shall show signal status of all intersections.
- The system shall display emergency alerts and congestion warnings.

6. Manual Control

- The admin shall manually override traffic signals when required.
- The admin shall control specific intersections directly.
- The admin shall reset traffic signals to automatic mode.

7. Alert Management

- The system shall generate congestion alerts.
- The system shall generate sensor failure alerts.
- The system shall generate emergency vehicle alerts.
- The system shall notify signal malfunction alerts.

8. Data Storage

- The system shall store traffic logs in the database.
- The system shall maintain historical traffic records.
- The system shall save emergency event history.

9. Reporting

- The system shall generate traffic performance reports.
- The system shall generate congestion analysis reports.
- The system shall generate emergency response reports.

10. User Authentication

- The system shall allow secure admin login.
- The system shall restrict unauthorized access.
- The system shall maintain admin activity logs.

5. NON- FUNTIONAL REQUIREMENTS

- **Performance:** System response time shall be less than **2 seconds**.
- **Reliability:** System uptime shall be **99.9%**.
- **Availability:** System shall operate **24/7** continuously.
- **Security:** Only authorized users can access the system.
- **Scalability:** System shall support multiple intersections.
- **Maintainability:** System shall allow easy updates and maintenance.
- **Accuracy:** Vehicle detection accuracy shall be above **95%**.
- **Usability:** Dashboard shall be simple and user-friendly.
- **Flexibility:** System shall support both automatic and manual control.
- **Data Integrity:** Traffic data shall remain accurate and secure.
- **Response Time:** Emergency override shall activate within **1 second**.
- **Compatibility:** System shall work with different sensors and traffic devices.

6. USE CASE DIAGRAM AND DESCRIPTON TABLE:

Actors

- **Traffic Sensors:** Collect real-time traffic data.
- **Vehicle:** Provides traffic flow information through movement.
- **Emergency Vehicle:** Sends emergency priority request.
- **Admin / Traffic Police:** Monitor and control system operations.

Main Use Cases

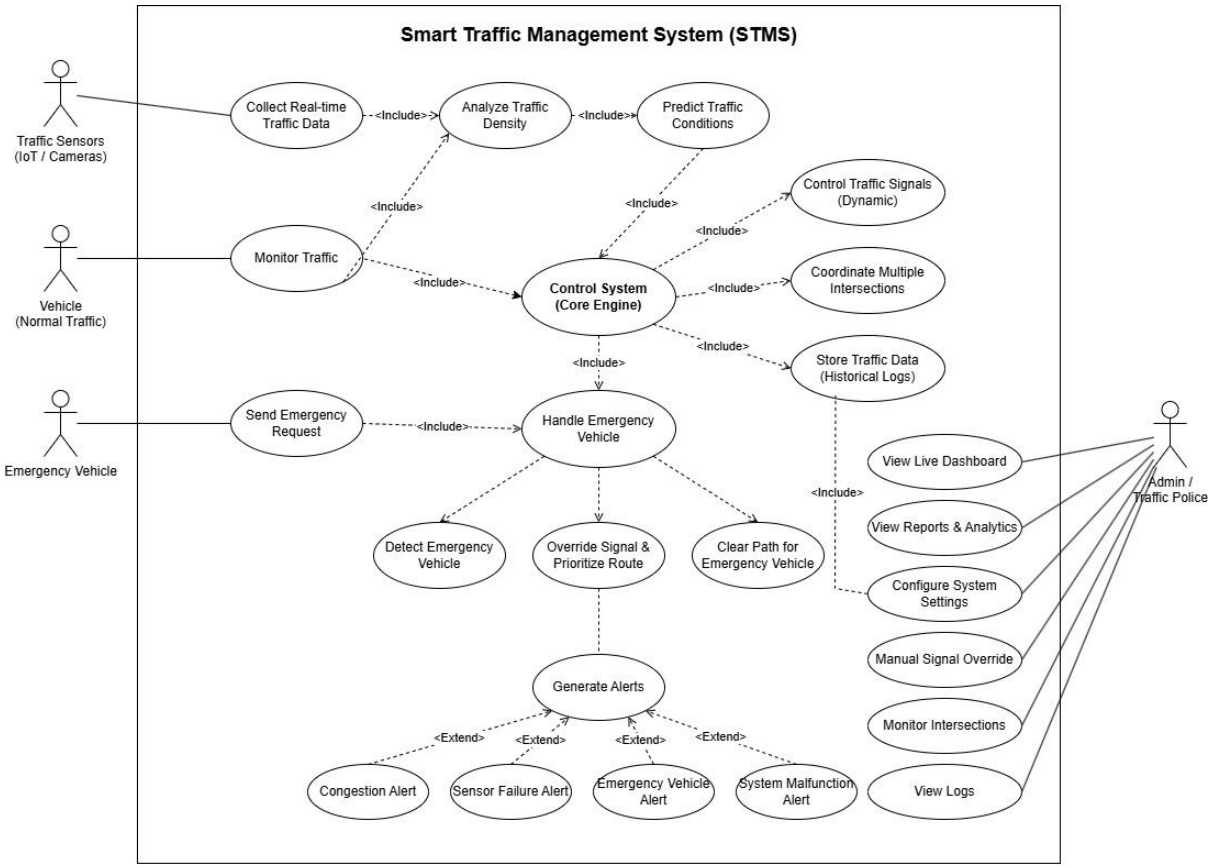
- **Collect Real-Time Traffic Data:** Sensors gather vehicle and road data.
- **Analyze Traffic Density:** System checks traffic congestion level.
- **Predict Traffic Conditions:** AI predicts future traffic flow.
- **Monitor Traffic:** System continuously observes traffic movement.
- **Control System (Core Engine):** Central unit that manages all decisions.
- **Control Traffic Signals:** Dynamically adjusts signal timing.
- **Coordinate Multiple Intersections:** Synchronizes signals across roads.
- **Store Traffic Data:** Saves traffic history and system logs.

Emergency Handling Use Cases

- **Send Emergency Request:** Emergency vehicle requests priority route.
- **Handle Emergency Vehicle:** System activates emergency processing.
- **Detect Emergency Vehicle:** Confirms emergency signal.
- **Override Signal & Prioritize Route:** Gives green path to emergency vehicle.
- **Clear Path for Emergency Vehicle:** Clears traffic route.
- **Generate Alerts:** Sends emergency and warning notifications

Admin Use Cases

- **View Live Dashboard:** Shows live traffic status.
- **View Reports & Analytics:** Displays traffic reports.
- **Configure System Settings:** Changes system parameters.
- **Manual Signal Override:** Admin manually controls signals.
- **Monitor Intersections:** Observe all intersections.
- **View Logs:** View stored activity records.



SMART TRAFFIC MANAGEMENT SYSTEM (STMS) — USE CASE DESCRIPTION TABLE

ACTOR	USE CASE ID	USE CASE NAME	DESCRIPTION	TRIGGER	MAIN FLOW (BASIC FLOW)	ALTERNATE FLOW / EXCEPTIONS	POST-CONDITION
Admin / Traffic Police	UC-01	Login	Admin logs into the system securely.	Admin opens login page.	• Enter username and password • System validates credentials • Dashboard is displayed	• Invalid credentials → Show error message	Admin is successfully authenticated.
	UC-02	View Dashboard	Admin views real-time traffic status and alerts.	Admin selects Dashboard.	• System retrieves live data • Dashboard shows traffic status, congestion, alerts	• Data not available → Show "No Data" message	Live traffic information is displayed.
	UC-03	Manual Signal Override	Admin manually changes traffic signal timings.	Admin selects Override option.	• Select intersection • Choose desired signal timing • Apply override	• Invalid timing → Show validation error	Signal timings are overridden.
	UC-04	Configure System	Admin configures system settings and parameters.	Admin selects Configuration.	• Modify settings (thresholds, timings, etc.) • Save configuration	• Invalid input → Show error message	New configuration is saved.
	UC-05	View Reports & Analytics	Admin views historical reports and analytics.	Admin selects Reports.	• Select report type and date range • System generates report • Display charts and insights	• No data available → Show message	Reports are generated and displayed.
Emergency Vehicle	UC-06	Request Priority Passage	Emergency vehicle requests priority to clear its route.	Emergency vehicle sends request.	• System receives emergency request • Captures location and route	• Request failed → Retry and notify dispatch	Request is received and acknowledged.
Traffic Sensors / Cameras	UC-07	Send Real-time	Sensors capture and send real-time traffic data.	Sensor detects vehicles.	• Detect vehicles and density	• Sensor failure → Alert system administrator	Traffic data is stored successfully.

		Traffic Data			• Send data to the system		
System (Automated)	UC-08	Monitor Traffic	System continuously monitors traffic conditions.	System timer / Data arrival.	• Receive data from sensors • Update current traffic status	• Data loss → Use last valid data and notify	Traffic is monitored in real time.
	UC-09	Collect Traffic Data	Collect data from all sensors and sources.	Triggered by Monitor Traffic.	• Aggregate data from all sensors • Validate and preprocess data	• Incomplete data → Log warning	Clean traffic data is available.
	UC-10	Analyze Traffic Data (AI)	AI engine analyzes traffic data for patterns.	After data collection.	• Run AI/ML model • Analyze density, speed, flow	• AI engine error → Use fallback logic	Traffic insights are generated.
	UC-11	Detect Traffic Condition	System detects current traffic condition.	After analysis.	• Evaluate congestion level • Classify: Normal / Congested / Emergency	• Uncertain condition → Mark as Unknown	Traffic condition is identified.
	UC-12	Adjust Signal Timing Dynamically	System adjusts signal timing automatically.	Triggered by condition detection.	• Calculate optimal timing • Send timing to signal controller	• Controller not reachable → Retry and alert	Signal timing is updated.
	UC-13	Coordinate Traffic Signals	System coordinates signals across intersections.	After timing adjustment.	• Synchronize neighboring signals • Optimize traffic flow	• Communication failure → Use local timing	Signals are synchronized.
	UC-14	Update Dashboard (Live Status)	System updates dashboard with latest status.	After data update.	• Push live data to dashboard • Update maps, charts, alerts	• Dashboard offline → Queue updates	Dashboard shows live status.
	UC-15	Store Data & Logs	System stores traffic data and logs.	Periodic / Event triggered.	• Save data to database • Maintain logs and history	• Database error → Retry and backup	Data is securely stored.
	UC-16	Generate Alerts & Notifications	System generates alerts for critical events.	When congestion or emergency detected.	• Create alert message • Send notifications to admin	• Notification failure → Retry via alternate channel	Alert is delivered successfully.

MILESTONE 2: MODELING

DATA FLOW DIAGRAM (DFD):

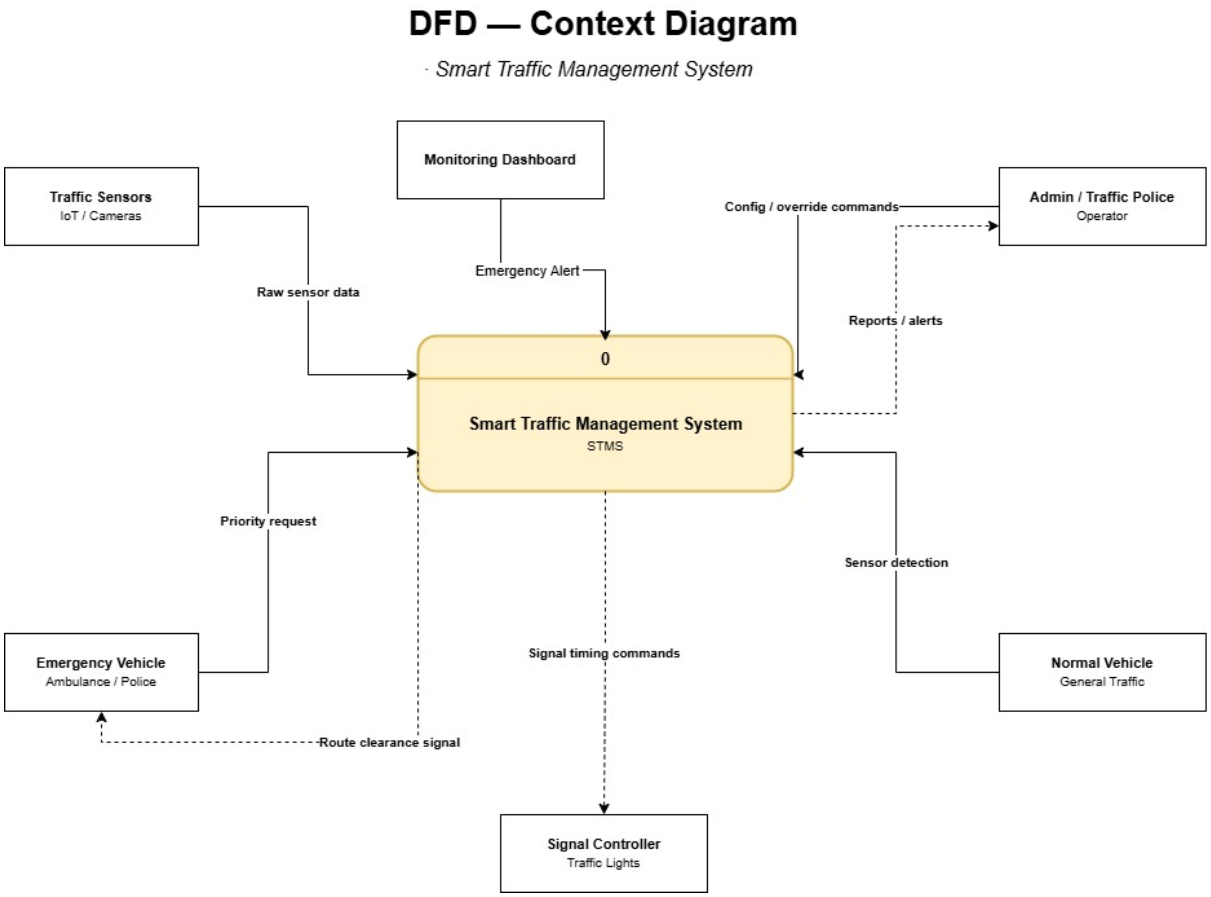
CONTEXT DAGRAM

External Entities:

- Vehicles
- Emergency Vehicles
- Sensors
- Admin
- Traffic Police

Central Process:

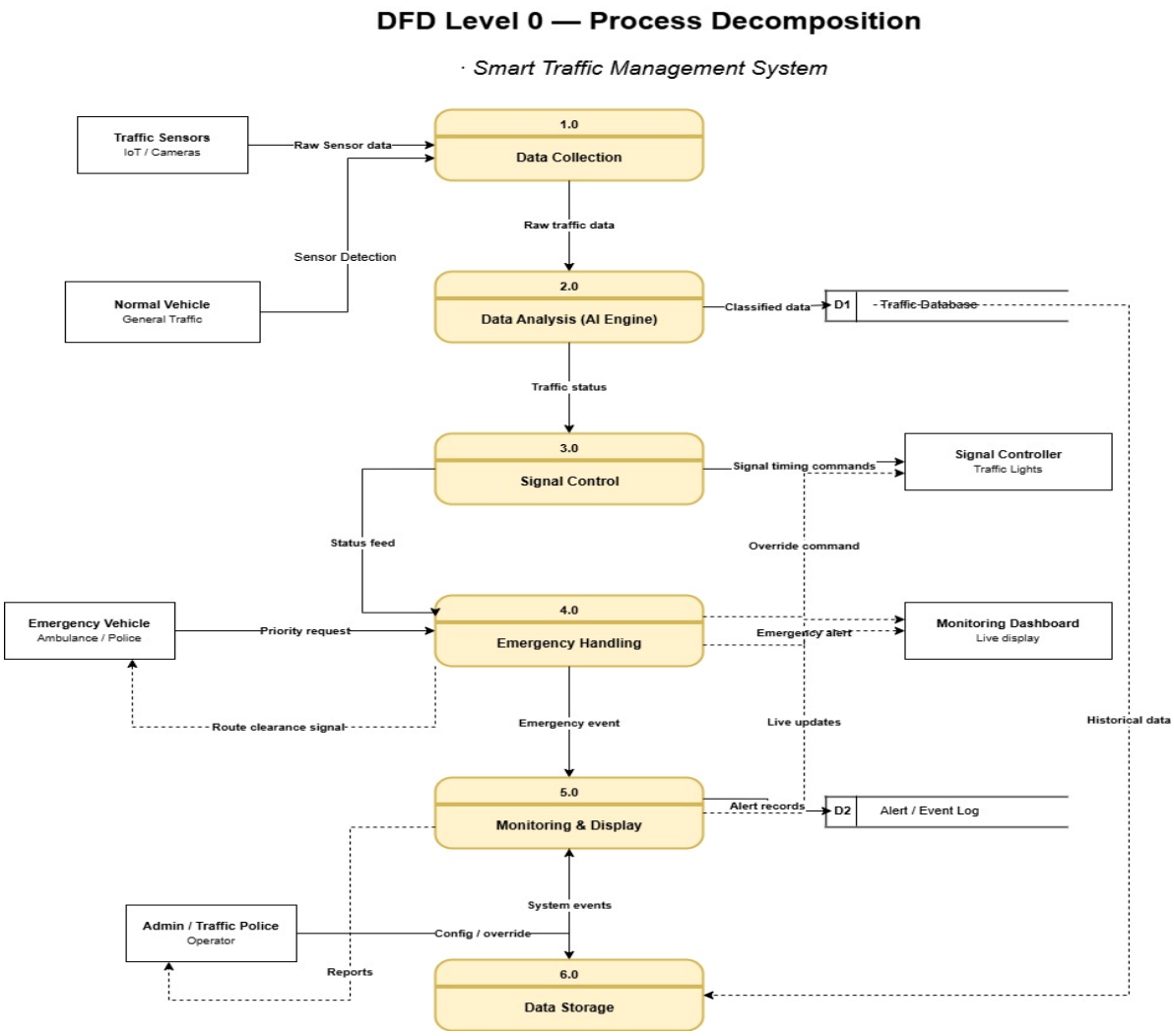
STMS



DFD LEVEL 0 DIAGRAM (Process Decomposition)

This diagram provides a high-level overview of the entire system's workflow:

- Data Entry: Traffic sensors and vehicles feed raw data into the Data Collection (1.0) process.
- AI Analysis: The Data Analysis (2.0) engine processes this information, categorizing it and storing it in a Traffic Database.
- Execution & Response: Based on the analyzed traffic status, the system generates Signal Control (3.0) commands for traffic lights.
- Special Cases: There is a dedicated Emergency Handling (4.0) process that manages priority requests from ambulances or police vehicles, allowing for signal overrides.
- Output & Logging: The system provides live updates to a Monitoring & Display (5.0) dashboard for administrators and stores all historical events and reports in Data Storage (6.0).



DFD LEVEL 1 DIAGRAM (Sub-Process Detail)

Expansion of P2 — Data Analysis (AI Engine)

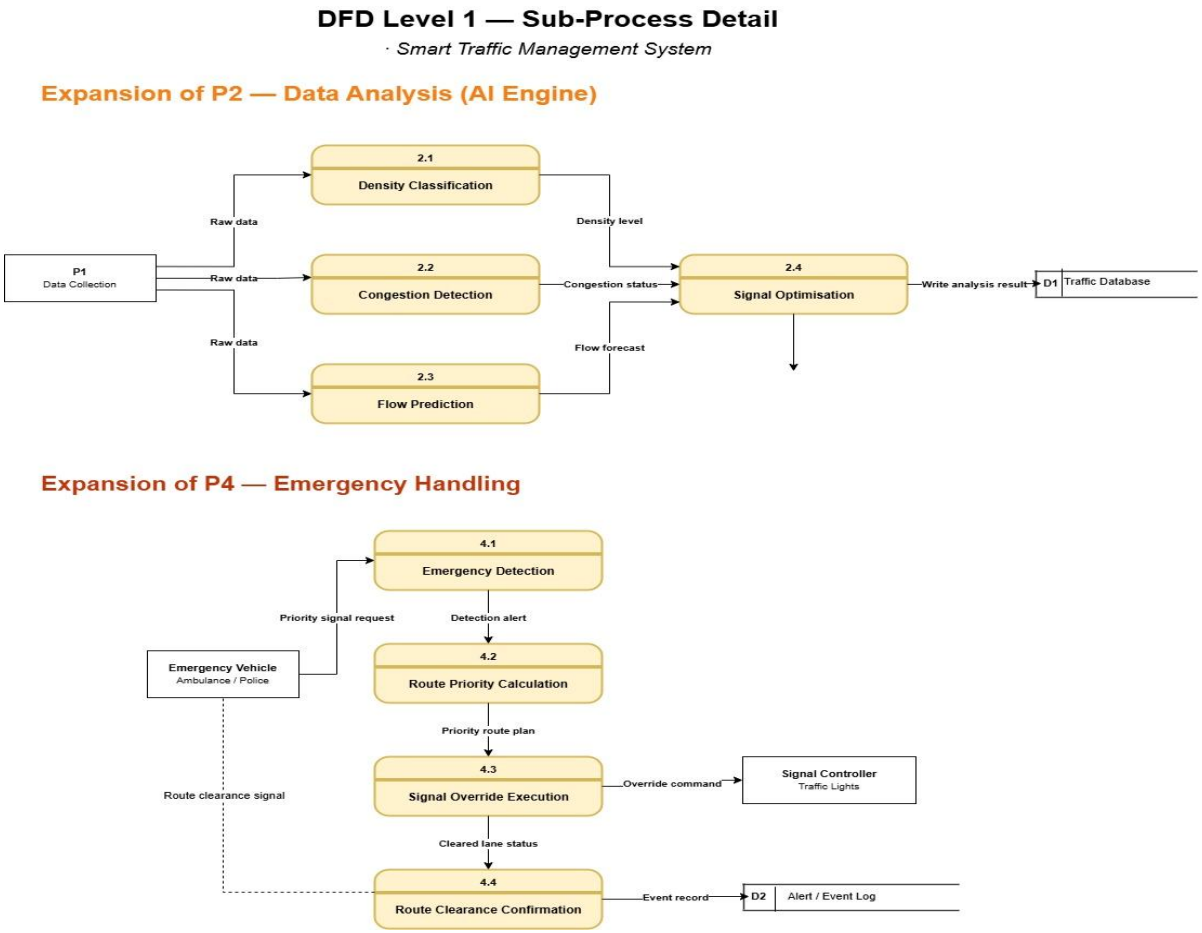
This breaks down how the AI evaluates traffic. It takes raw data from the collection phase and splits it into three parallel analytical tasks:

- Density Classification (2.1): Determining how crowded the road is.
- Congestion Detection (2.2): Identifying specific traffic jams.
- Flow Prediction (2.3): Forecasting future traffic patterns.
- All three inputs feed into Signal Optimisation (2.4), which writes the results to the database to inform light timing.

Expansion of P4 — Emergency Handling

This details the life cycle of an emergency vehicle's passage through the system:

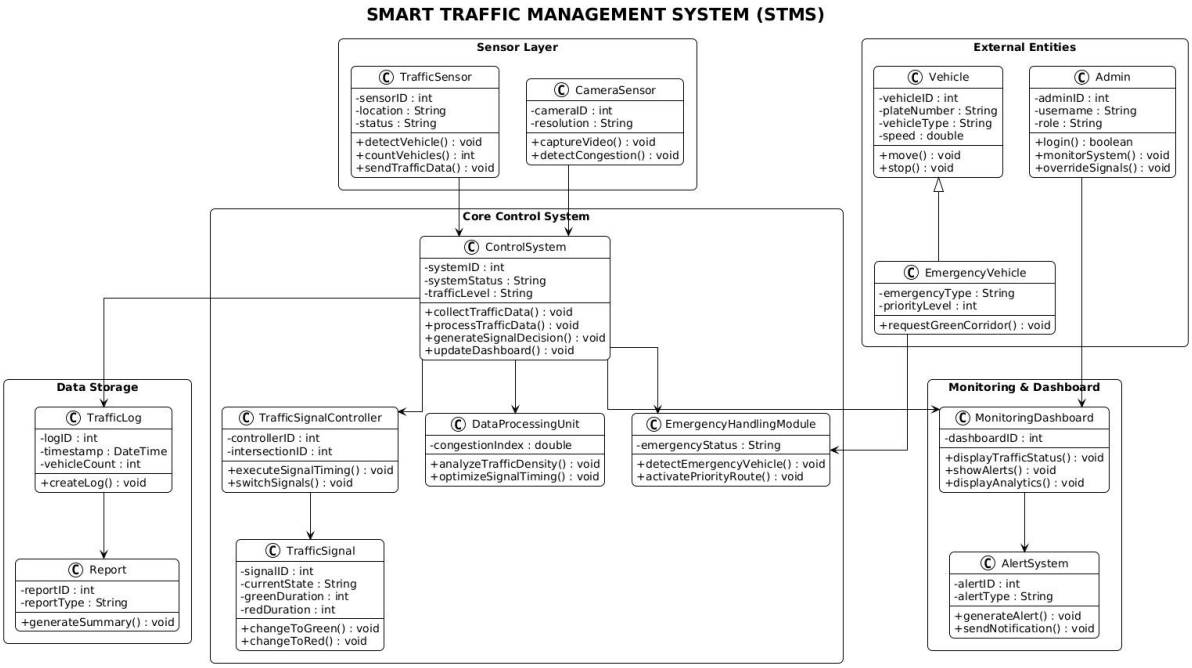
- Detection: It identifies a priority signal request from an emergency vehicle (4.1).
- Planning: It calculates the most efficient route priority (4.2).
- Action: It sends an immediate Override Command to the traffic light controllers (4.3) to clear the path.
- Verification: Finally, it confirms the lane is cleared and logs the event for administrative records (4.4).



CLASS DIAGRAM:

Data Structures & Logic

- **Sensor Layer**: Defines the TrafficSensor and CameraSensor classes, which are responsible for detectVehicle() and detectCongestion() operations to feed raw inputs into the system.
- **Core Control System**: This is the primary logic tier where the ControlSystem class coordinates generateSignalDecision() and updateDashboard() tasks. It utilizes the DataProcessingUnit to calculate a congestionIndex and the EmergencyHandlingModule to manage emergencyStatus.
- **Signal Execution**: The TrafficSignalController class translates logical decisions into physical actions via the TrafficSignal class, which manages state variables like greenDuration and redDuration.



SEQUENCE DIAGRAM:

Normal Traffic Monitoring

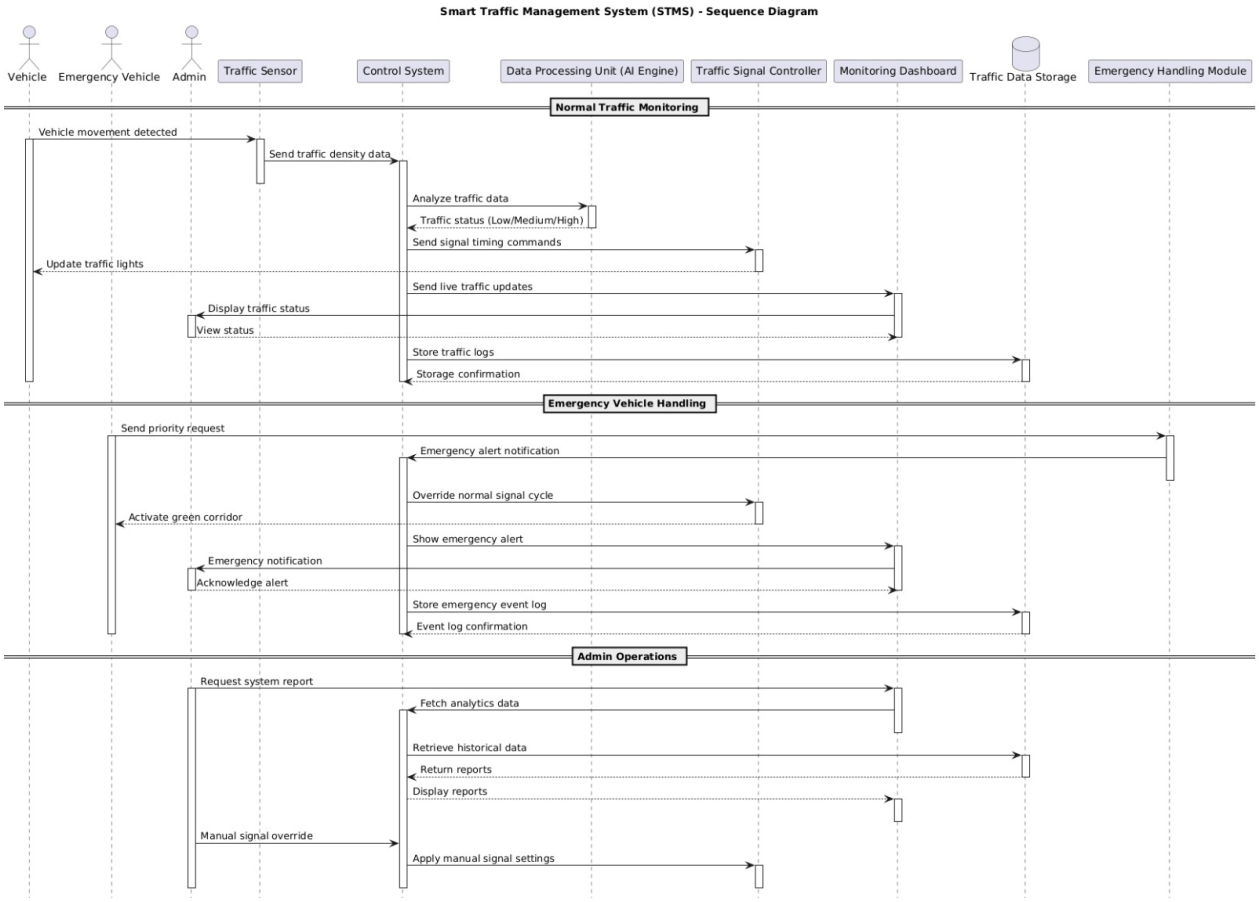
- Detection: A Traffic Sensor detects vehicle movement and sends density data to the Control System.
- Analysis & Action: The Data Processing Unit (AI Engine) categorizes traffic status (Low/Medium/High), allowing the Control System to send timing commands to the Traffic Signal Controller.
- Visibility & Logging: Live updates are pushed to the Monitoring Dashboard for the Admin, while traffic logs are archived in Traffic Data Storage.

Emergency Vehicle Handling

- Priority Trigger: An Emergency Vehicle sends a priority request directly to the Emergency Handling Module.
- System Override: The system issues an emergency alert notification, overrides the normal signal cycle, and activates a "green corridor" for the vehicle.
- Alerting: The Admin is notified of the emergency on the dashboard, and the event is logged for legal and system records.

Admin Operations

- Reporting: The Admin can request system reports, which triggers the Control System to fetch analytics and historical data from storage to display on the dashboard.
- Manual Control: The Admin has the authority to perform a Manual signal override, sending specific signal settings directly to the Traffic Signal Controller



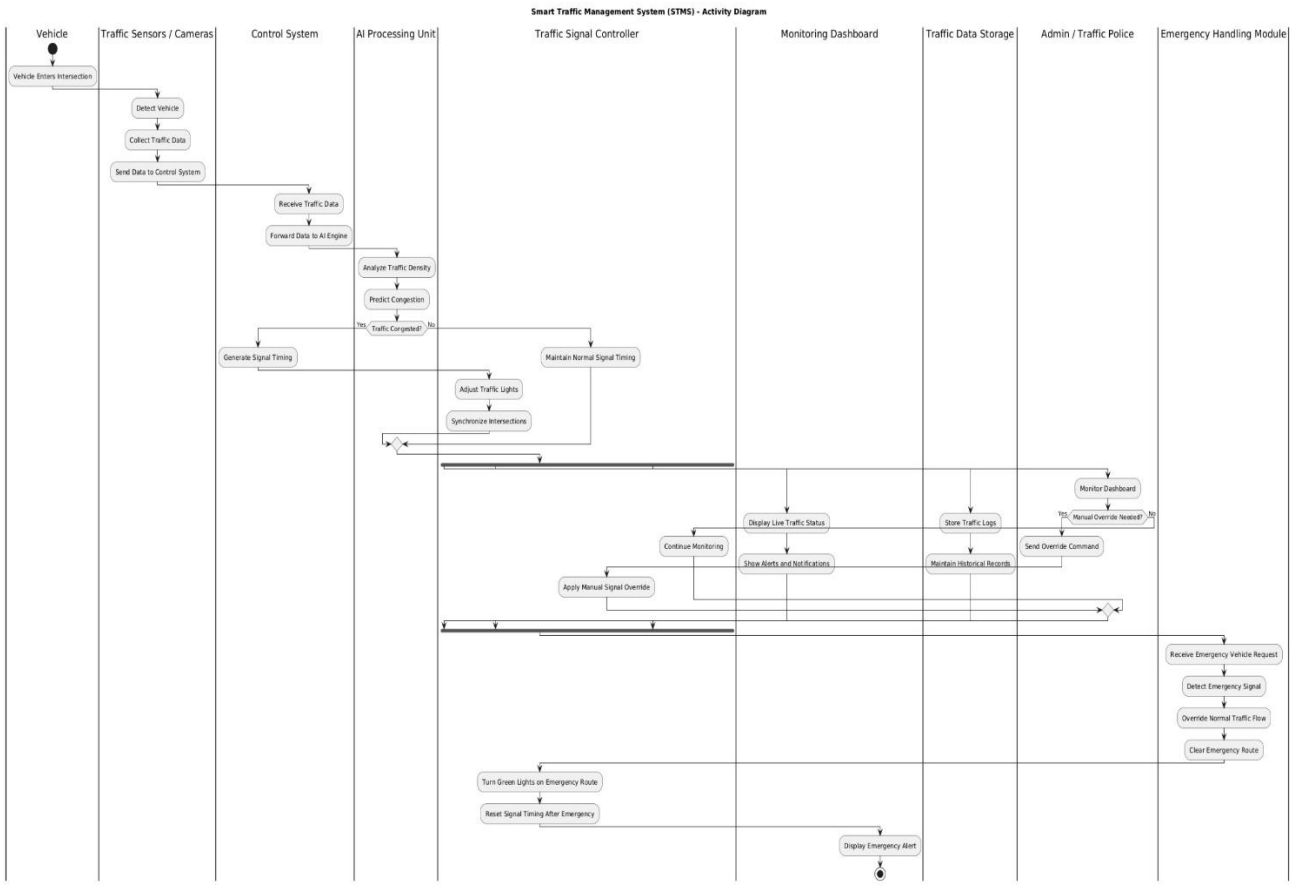
ACTIVITY DIAGRAM:

Traffic Detection & Analysis: Sensors detect vehicles and send data to the Control System, which forwards it to the **AI Processing Unit** to analyze density and predict congestion.

Signal Optimization: If traffic is congested, the system generates new timings to adjust lights and synchronize intersections; otherwise, it maintains normal timing.

Monitoring & Logging: The system simultaneously displays live status on the **Monitoring Dashboard**, stores traffic logs, and allows **Admin/Traffic Police** to send manual overrides if needed.

Emergency Response: The **Emergency Handling Module** can bypass normal flow to clear routes, which results in the Traffic Signal Controller turning lights green along the emergency path.



STATE DIAGRAM:

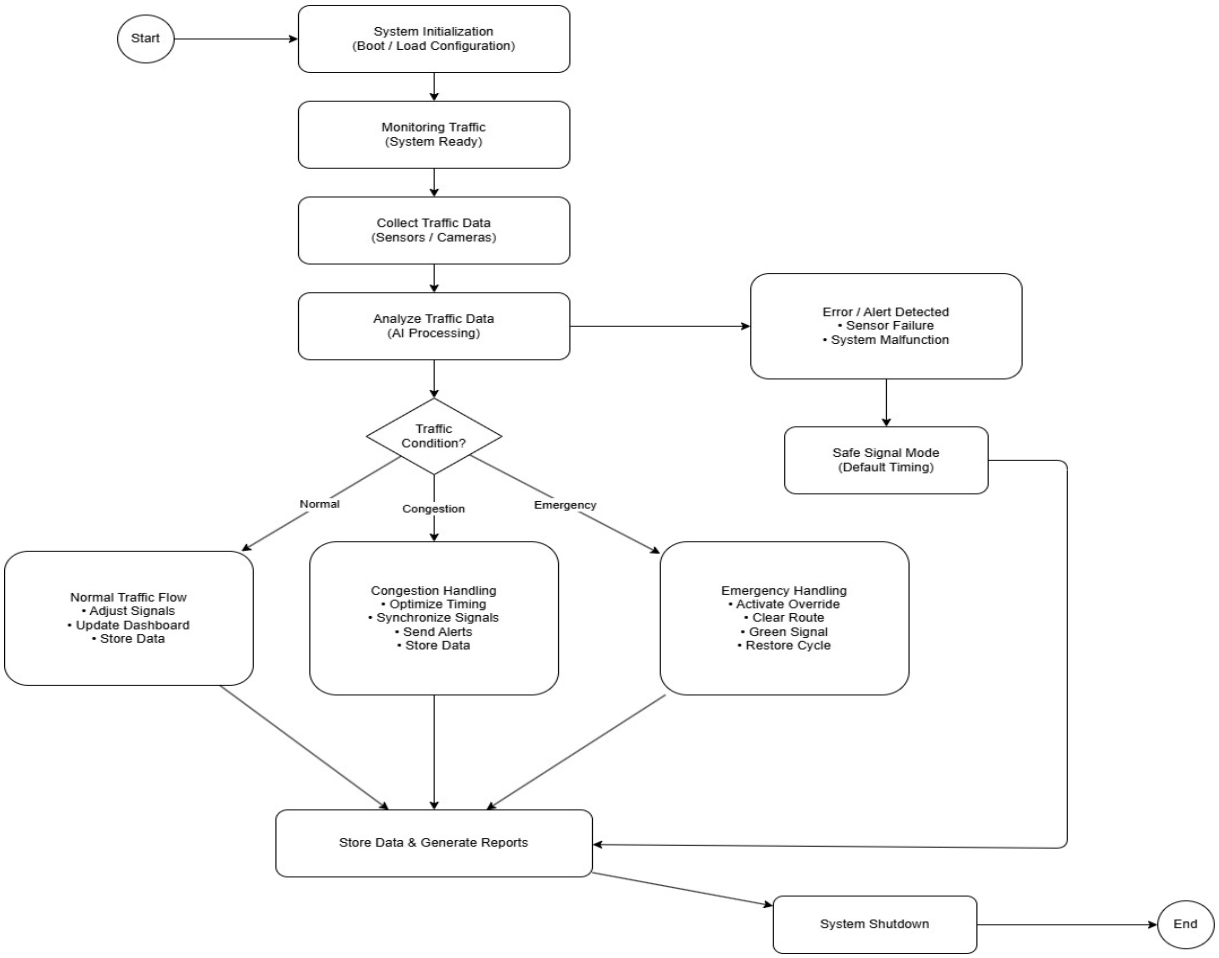
- System **Initialization**: Boots the system and loads necessary configurations.

- Monitoring **Traffic**: The "System Ready" state where the environment is actively scanned.

- Collect **Traffic Data**: Gathers raw information using sensors and cameras.
- Analyse **Traffic Data**: Uses AI processing to evaluate current road conditions.
- Normal **Traffic Flow**: Standard operation that adjusts signals and updates the dashboard.
- Congestion **Handling**: Optimizes and synchronizes signals to resolve heavy traffic.
- Emergency **Handling**: Activates overrides to clear routes for priority vehicles.
- Error / **Alert Detected**: A fail-state triggered by sensor failure or system malfunctions.
- Safe **Signal Mode**: A backup state that runs default timing to ensure basic safety.
- Store **Data & Generate Reports**: Finalizes records and archives system performance data.
- System **Shutdown**: Safely deactivates all processes to end operations.

State Diagram

· Smart Traffic Management System



MILESTONE 3 — DESIGN & ARCHITECTURE

Refined UML Diagrams

1. Refined Use Case Diagram

The Use Case Diagram was refined to improve clarity and system interaction accuracy.

Refinements Made:

- Added Admin as the primary controlling actor.
- Added Traffic Sensor as a secondary actor for data input.
- Added Emergency Vehicle as an external triggering actor.
- Improved use case relationships using <<include>> and <<extend>>.
- Connected Monitor Traffic with Analyze Traffic Density using include relation.
- Connected Emergency Priority Handling with Control Signal Timing using extend relation.
- Added Generate Reports as admin-specific functionality.
- Improved system boundary definition for better modular understanding.
- Removed redundant actor interactions for simplicity.

2. Refined Activity Diagram

The Activity Diagram was refined to improve process flow and decision clarity.

Refinements Made:

- Added clear start and end nodes.
- Improved traffic detection flow sequence.
- Added decision node for Traffic Density Check.
- Added separate decision branch for Emergency Vehicle Detection.
- Refined congestion handling process.
- Added loop for continuous monitoring.
- Improved signal adjustment logic.
- Added fallback path for normal traffic cycle.
- Removed unnecessary process duplication.

3. Refined Sequence Diagram

The Sequence Diagram was refined to improve interaction sequence accuracy.

Refinements Made:

- Clearly defined lifelines for:
  - Emergency Vehicle
  - Traffic Sensor
  - Control System
  - Traffic Signal

- Improved message ordering between entities.
- Added activation bars for system processing.
- Added emergency detection message trigger.
- Added priority signal activation response.
- Added system return-to-normal operation flow.
- Improved synchronization of message passing.
- Removed redundant message calls.
- Improved response timing sequence for better logical consistency.

Layered Architecture

Layer 1: Presentation Layer

Handles:

- Dashboard
- Alerts
- Reports
- Admin interface

Layer 2: Application Layer

Handles:

- Business logic
- Request processing
- Rule execution

Layer 3: Processing Layer

Handles:

- AI traffic analysis
- Prediction engine
- Signal optimization

Layer 4: Data Layer

Handles:

- Database
- Logs
- Historical records

Layer 5: Hardware Layer

Handles:

- Cameras
- Sensors
- Signal devices

Why this architecture?

Because traffic systems require:

- Separation of duties
- Real-time processing
- Hardware interaction
- Reliability
- Easy upgrades

MILESTONE 4 — TESTING & MANAGEMENT

UNIT TEST CASES

Test Case ID	Test Name	Input	Expected Output	Actual Result
TC-001	Vehicle Detection	Vehicle enters lane	Vehicle detected	Pass
TC-002	Density Calculation	50 vehicles detected	Density = High	Pass
TC-003	Signal Timing Update	High congestion	Green extended to 90 sec	Pass
TC-004	Low Traffic Timing	Low traffic	Green reduced to 30 sec	Pass
TC-005	Emergency Detection	Ambulance signal	Emergency mode ON	Pass
TC-006	Route Clearance	Emergency active	Clear traffic route	Pass
TC-007	Dashboard Update	Traffic data change	Live dashboard refresh	Pass
TC-008	Manual Override	Admin changes signal	Signal updated	Pass
TC-009	Invalid Login	Wrong credentials	Access denied	Pass
TC-010	Data Storage	Traffic event completed	Saved to database	Pass

Test Case ID	Test Name	Input	Expected Output	Actual Result
TC-011	Sensor Failure Alert	Sensor offline	Alert generated	Pass
TC-012	Report Generation	Daily request	PDF report generated	Pass

INTEGRATION TESTS

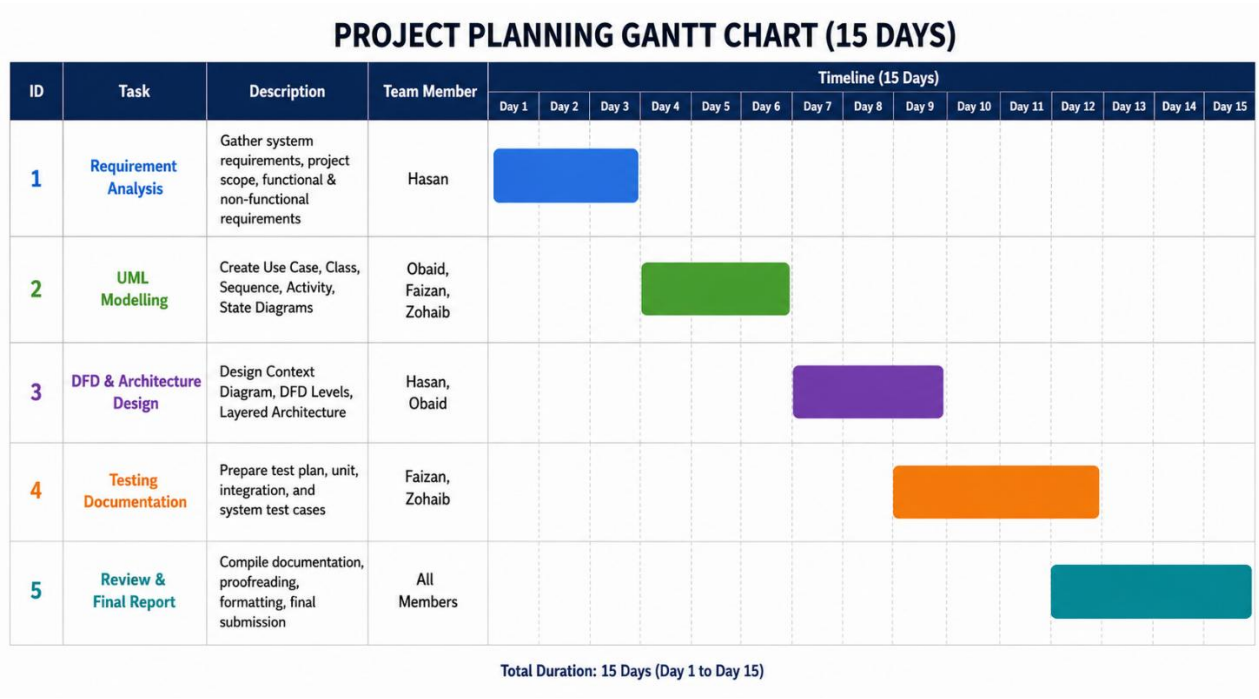
Test Case ID	Module Integration	Expected Output	Result
IT-01	Sensor + AI Engine	Correct analysis	Pass
IT-02	AI Engine + Signal Controller	Signal updates correctly	Pass
IT-03	Emergency Module + Dashboard	Emergency shown live	Pass

SYSTEM TESTS

Test Case ID	Scenario	Expected Output	Result
ST-01	Peak Hour Simulation	Traffic optimized	Pass
ST-02	Emergency During Congestion	Priority route cleared	Pass
ST-03	Multi-Intersection Control	All signals synchronized	Pass

Project Planning Chart (Gantt Chart)

Week	Task	Description	Team Member
Week 1	Requirement Analysis	Gather system requirements, project scope, functional & non-functional requirements	Hasan
Week 1	UML Modelling	Create Use Case, Class, Sequence, Activity, State Diagrams	Obaid, Faizan, Zohaib
Week 2	DFD & Architecture Design	Design Context Diagram, DFD Levels, Layered Architecture	Hasan, Obaid
Week 2	Testing Documentation	Prepare test plan, unit, integration, and system test cases	Faizan, Zohaib, Hasan
Week 2	Review & Final Report	Compile documentation, proofreading, formatting, final submission	All Members



Risk Analysis Table

Risk ID	Risk Description	Probability	Impact	Risk Level	Solution / Mitigation
R1	Sensor Failure	High	High	Critical	Install backup sensors
R2	Network Failure	Medium	High	High	Use local offline control mode
R3	Emergency Signal Detection Failure	Low	Very High	Critical	Add signal verification system
R4	Data Loss / Database Failure	Medium	High	High	Daily database backup
R5	Signal Controller Malfunction	Medium	High	High	Manual override by admin
R6	Unauthorized Access	Low	High	Medium	Secure login authentication
R7	System Software Crash	Medium	High	High	Automatic restart mechanism
R8	Incorrect Traffic Prediction	Medium	Medium	Medium	Improve AI training model
R9	Delayed Alert Generation	Low	Medium	Medium	Optimize processing speed
R10	Hardware Maintenance Delay	Medium	Medium	Medium	Regular maintenance schedule

Risk Management Strategy

- **Avoid:** Prevent system failures before they happen
- **Reduce:** Use backup systems and maintenance
- **Transfer:** Assign technical risks to maintenance teams
- **Accept:** Minor risks with low impact can be monitored only

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2. Software Engineering: A Practitioner's Approach
   McGraw-Hill Education
   https://www.mheducation.com
3. OMG UML Specification
   Unified Modelling Language standard documentation
4. IEEE Software Engineering Standards
   Software engineering documentation standards
5. ScienceDirect Intelligent Traffic Management Research
   Research papers on smart traffic management systems

Tools Used for Smart Traffic Management System Project

1. Draw.io (Diagrams.net)

Used for creating:

- Use Case Diagram
- Class Diagram
- Sequence Diagram
- Activity Diagram
- State Machine Diagram
- DFD Diagrams
- Architecture Diagrams

Purpose: Visual system modelling and design.

2. ChatGPT

Used for:

- Requirement analysis

- Functional and non-functional requirements writing
- Documentation assistance
- Test case generation
- Project planning support

Purpose: Documentation and system analysis support.

3. Claude AI

Used for:

- Research assistance
- Refining software engineering explanations
- Improving technical documentation

Purpose: Supporting analysis and documentation improvement.

4. Microsoft Word

Used for:

- Writing final project documentation
- Formatting assignment report
- Organizing diagrams and reports

Purpose: Final report preparation.